Large-Angle Statistics of the CMB Angular Correlation Function: Testing ΛCDM at Low Multipoles

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Abstract

We investigate the statistical properties of the cosmic microwave background (CMB) angular correlation function at large angular scales ($\ell \leq 30$) using Planck legacy data. Three complementary statistics are studied: the integrated squared correlation S_a^b, the interval-averaged correlation $\langle\xi\rangle_a^b$, and the antipodal correlation C_{180}. Theoretical distributions are generated via two independent methods — sampling from the MCMC posterior of the standard flat ΛCDM model fitted with the CamSpec high-multipole TT likelihood, and Monte Carlo realizations around the best-fit spectrum — allowing a direct comparison of the two approaches. We report the empirical p-values for each statistic and discuss the apparent sensitivity of the inferred tensions to the choice of theoretical ensemble.

Keywords: cosmic microwave background, angular correlation function, large-angle anomalies, Planck, ΛCDM

1. Introduction

The cosmic microwave background (CMB) provides our cleanest window onto the primordial universe. On the largest angular scales — corresponding to multipoles $\ell \lesssim 30$ — the temperature anisotropy pattern carries direct information about the very largest coherent structures in the observable universe. The standard ΛCDM model predicts that the fluctuations should be a nearly Gaussian, statistically isotropic random field fully characterized by the angular power spectrum C_ℓ.

Several anomalies have been reported at low multipoles since the first-year WMAP data release and confirmed in subsequent analyses of the Planck temperature maps (Aghanim et al., 2020). Among the most discussed is the apparent lack of large-angle correlations: the two-point angular correlation function $C(\theta)$ is observed to be anomalously small for $\theta \gtrsim 60°$, a feature quantified by the statistic $S_{1/2}$ (Spergel et al., 2003; Copi et al., 2013). While cosmic variance is large at these scales and makes rigorous significance assessment difficult, the tension with ΛCDM predictions deserves careful scrutiny.

In this work we re-examine these statistics using Planck legacy chains and two complementary methods to construct the theoretical sampling distribution. We focus on three statistics defined over five angular intervals that partition the accessible range $[0°, 180°]$: the quadratic statistic S_a^b, the linear statistic $\langle\xi\rangle_a^b$, and the antipodal correlation C_{180}. By comparing distributions generated from MCMC posterior samples to those obtained from Monte Carlo realizations around the best-fit spectrum, we probe the dependence of the inferred tensions on the choice of ensemble.

The primary model considered throughout is the standard flat ΛCDM fit to the Planck temperature power spectrum using the CamSpec high-multipole TT likelihood combined with the low-ℓ temperature and polarization likelihoods (Aghanim et al., 2020). Results for the Plik-based likelihood and for models with free curvature Ω_k are reported in the appendix.

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2. Statistics

2.1. Angular Power Spectrum and Correlation Function

The CMB temperature anisotropy $\Delta T(\hat{n})/T_0$ is expanded in spherical harmonics with coefficients $a_{\ell m}$. For a statistically isotropic Gaussian field the two-point statistics are fully encoded in the angular power spectrum $C_\ell = \langle |a_{\ell m}|^2 \rangle$. We work with the conventional scaled spectrum $D_\ell = \ell(\ell+1)C_\ell/(2\pi)$, which is approximately flat for a scale-invariant primordial spectrum.

The angular two-point correlation function is obtained by transforming back to real space via a Legendre expansion,

$$C(\theta) = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos\theta) = \sum_\ell \frac{2\ell+1}{2\ell(\ell+1)} D_\ell P_\ell(\cos\theta), \tag{1}$$

where P_ℓ are Legendre polynomials. The sum runs from $\ell = 2$ to a chosen $\ell_{\max} = 30$ throughout this work.

2.2. Interval Statistics

We study three statistics on each of the angular intervals listed in Table~1.

Antipodal correlation $C_{180} \equiv C(\theta = 180°)$ is the correlation between antipodal points on the sky and is relatively insensitive to small-scale masking effects.

Interval-averaged correlation $\langle\xi\rangle_a^b$ is the mean of $C(\theta)$ over an angular interval $[\theta_a, \theta_b]$, equivalently expressed in terms of $\mu = \cos\theta$ as

$$\langle\xi\rangle_a^b = \frac{1}{b-a}\int_a^b C(\theta)\, d\mu = \sum_\ell \frac{D_\ell}{2\ell(\ell+1)} \frac{P_{\ell+1}(b) - P_{\ell-1}(b) - P_{\ell+1}(a) + P_{\ell-1}(a)}{b-a}, \tag{2}$$

where $a = \cos\theta_b$ and $b = \cos\theta_a$. Being linear in D_ℓ, this statistic is sensitive to shifts in the mean correlation level.

Integrated squared correlation S_a^b is defined as

Table 1: Angular intervals used for S_a^b and $\langle\xi\rangle_a^b$.

Angular range	Description	$[\cos\theta_b,\ \cos\theta_a]$
$[0°,\ 30°]$	Small angles	$[0.865,\ 1.000]$
$[30°,\ 92°]$	Intermediate	$[-0.039,\ 0.865]$
$[92°,\ 154°]$	Large angles	$[-0.901,\ -0.039]$
$[154°,\ 180°]$	Near-antipodal	$[-1.000,\ -0.901]$
$[60°,\ 180°]$	Classic $S_{1/2}$	$[-1.000,\ 0.500]$

$$S_a^b = \int_a^b C^2(\theta)\, \sin\theta\, d\theta = \sum_{n,m} A_n A_m D_n D_m T_{nm}, \tag{3}$$

where $A_\ell = (2\ell+1)/[2\ell(\ell+1)]$ and T_{nm} are overlap integrals of Legendre polynomials over the interval. The classic $S_{1/2}$ statistic corresponds to the interval $[60°, 180°]$ (Spergel et al., 2003). Being quadratic in D_ℓ, this statistic amplifies deviations that are coherent across multipoles.

3. Data

3.1. Planck Posterior Chains

We use the publicly available Planck 2018 MCMC chains from the Planck Legacy Archive[1]. The primary dataset is the CamSpec high-multipole TT likelihood combined with the low-ℓ temperature and polarization likelihoods (`base_CamSpecHM_TT_lowl_lowE`). This provides posterior samples of the six standard ΛCDM parameters $\{\Omega_b h^2,\ \Omega_c h^2,\ H_0,\ \tau,\ n_s,\ A_s\}$ along with a suite of derived quantities. The Plik-based TT likelihood and two curvature-extended runs are considered in Appendix~Appendix A.

For each posterior sample the theoretical power spectrum D_ℓ is computed with CAMB (Lewis), and the corresponding $C(\theta)$ and derived statistics are evaluated analytically using equations 1–3.

3.2. Observed Planck Multipoles

The experimental values of each statistic are computed from the Planck 2018 Commander temperature map and associated beam and pixel window functions, using the same multipole range $\ell \in [2, 30]$.

[1] https://pla.esac.esa.int/

These observed values serve as the reference against which the theoretical distributions are compared.

P-values are defined empirically: the fraction of Monte Carlo or MCMC-derived realizations whose statistic is at least as extreme (in absolute deviation from the median) as the observed value. The multiple statistics studied here are correlated, so individual p-values should not be combined naively.

3.3. Ensemble Construction

Two independent strategies are used to construct the theoretical sampling distribution:

1. **MCMC ensemble.** The last 1000 samples from the Planck posterior chain are converted to CAMB spectra, where one realization is generated for each sample. The resulting set of 1000 theoretical D_ℓ vectors spans the parameter uncertainty and constitutes a Bayesian predictive distribution for each statistic. In this approach we directly propagate the full posterior uncertainty through to the derived statistics, and we also take in to account the cosmic variance for each sample
2. **Best-fit ensemble.** The single best-fit parameter vector is used to compute D_ℓ^{bf}. An ensemble of 1000 realizations is then obtained by drawing independent samples of each $a_{\ell m}$ from the chi-squared distribution dictated by C_ℓ^{bf} and reconstructing D_ℓ from those coefficients. This approach captures cosmic-variance scatter around the best-fit spectrum while fixing the cosmological parameters.

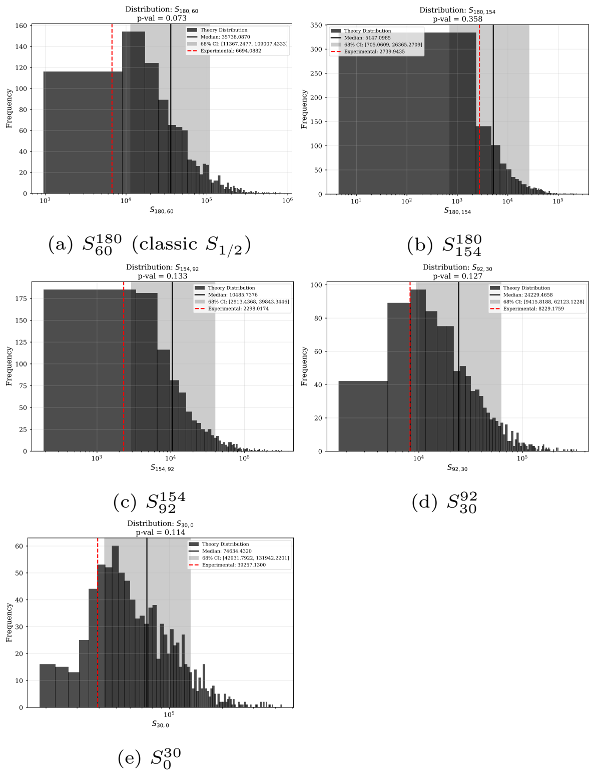
In both cases the realization where made using the (Sullivan et al., 2024) code, which takes in to account masking effects and the Planck beam and pixel window functions. The resulting D_ℓ are then transformed to $C(\theta)$ and the derived statistics using the same analytical expressions as for the MCMC samples.

All the code can be found in the GitHub repository here. With a small sample of the data and the code, so the analysis can be extended to more angle intervals or different models with a Planck format (ie. using the same names for the aparemeters).

4. Results: Best-Fit Analysis

In the best-fit analysis the theoretical sampling distribution is generated by Monte Carlo realizations of the CMB sky drawn from the best-fit CamSpec ΛCDM spectrum. This approach isolates cosmic variance from parameter uncertainty.

Figures 1 and 2 display the distributions of the S_a^b and $\langle\xi\rangle_a^b$ statistics respectively, for all five angular intervals. Each panel shows the distribution over the 1000 Monte Carlo realizations (blue histogram), and the observed Planck value (red vertical line). Figure~3 shows the analogous distribution for C_{180}.



(a) S_{60}^{180} (classic $S_{1/2}$)
(b) S_{154}^{180}
(c) S_{92}^{154}
(d) S_{30}^{92}
(e) S_{0}^{30}

Figure 1: Best-fit Monte Carlo distributions of the S_a^b statistic for model `base_CamSpecHM_TT_lowl_lowE` ($\ell_{\max} = 30$). Each panel shows the normalized histogram of 1000 realizations; the red line marks the observed Planck value.

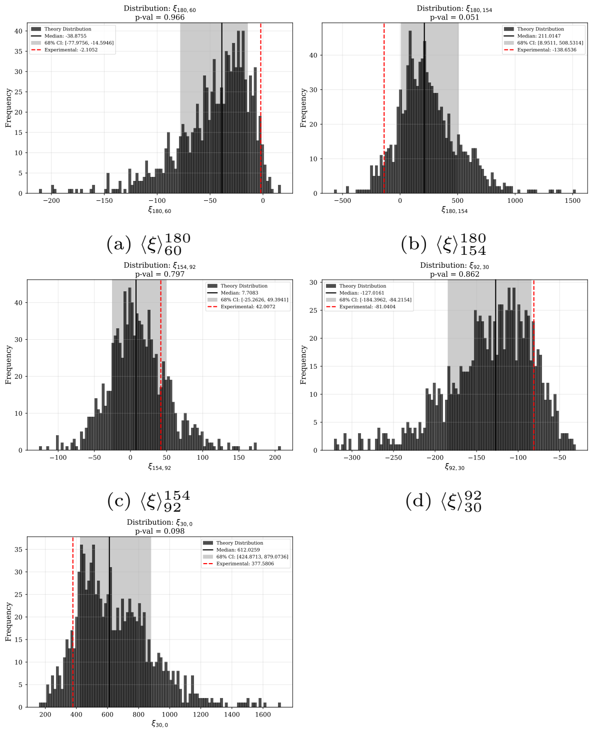
The statistical summary — including the median, 68% credible interval, observed value, and empirical p-value for each statistic — is reported in Table~2.

5. Results: MCMC Analysis

In the MCMC analysis the theoretical sampling distribution is constructed directly from the Planck posterior, propagating full parameter uncertainty through to the derived statistics. The 1000 posterior samples drawn from the CamSpec ΛCDM chain

Table 2: Statistical summary for the best-fit analysis of model `base_CamSpecHM_TT_lowl_lowE`. Columns give the experimental value, best-fit median, 68% interval, and empirical p-value.

Statistic	Experimental	MCMC Median	68% CI	p-value
C_{180}	-354.1497 ± 171.2834	260.7976	$[-38.2572, 629.7045]$	0.022
$S_{180,60}$	6694.0882 ± 8822.2915	35725.0265	$[11308.2691, 108785.4640]$	0.073
$S_{180,154}$	2739.9435 ± 3018.8131	5144.2773	$[704.4897, 26362.7403]$	0.358
$S_{154,92}$	2298.0174 ± 3103.8229	10463.2839	$[2893.9033, 39807.8958]$	0.133
$S_{92,30}$	8229.1759 ± 10643.1270	24227.7116	$[9406.5535, 62090.4792]$	0.127
$S_{30,0}$	$39257.1300 \pm 48919.8333$	74487.4419	$[42884.9866, 131879.2429]$	0.114
$\xi_{180,60}$	-2.1052 ± 12.1260	-38.8758	$[-78.4081, -14.6241]$	0.966
$\xi_{180,154}$	-138.6536 ± 57.5605	210.8664	$[7.7966, 508.4919]$	0.051
$\xi_{154,92}$	42.0072 ± 34.6449	7.6378	$[-25.2907, 49.3801]$	0.797
$\xi_{92,30}$	-81.0404 ± 79.0240	-127.2322	$[-184.4908, -84.2408]$	0.862
$\xi_{30,0}$	377.5806 ± 351.5645	611.8525	$[424.7966, 878.9550]$	0.098



(a) $\langle\xi\rangle^{180}_{60}$ (b) $\langle\xi\rangle^{180}_{154}$ (c) $\langle\xi\rangle^{154}_{92}$ (d) $\langle\xi\rangle^{92}_{30}$ (e) $\langle\xi\rangle^{30}_{0}$

Figure 2: Best-fit Monte Carlo distributions of the interval-averaged correlation $\langle\xi\rangle^b_a$ for model `base_CamSpecHM_TT_lowl_lowE`.

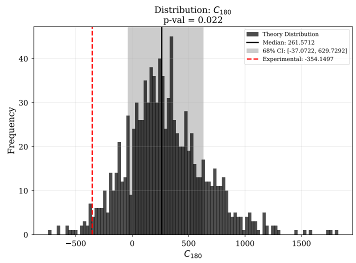


Figure 3: Best-fit Monte Carlo distribution of the antipodal correlation C_{180} for model `base_CamSpecHM_TT_lowl_lowE`.

each yield a distinct D_ℓ spectrum, whose distribution reflects both cosmic variance and parameter degeneracies.

Figures 4, 5, and 6 show the resulting distributions of S^b_a, $\langle\xi\rangle^b_a$, and C_{180} respectively. The broader width of the MCMC-derived histograms compared to the best-fit case reflects the additional spread due to cosmological parameter uncertainty.

The quantitative results for the MCMC ensemble are collected in Table~3.

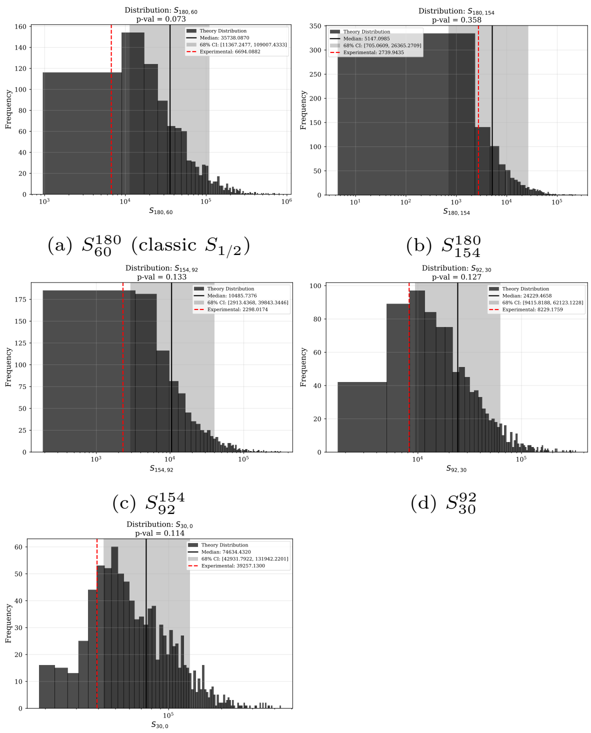
6. Discussion

The two ensembles — best-fit Monte Carlo and MCMC posterior — provide complementary perspectives on the significance of the large-angle anomalies.

The best-fit ensemble captures cosmic-variance scatter at fixed cosmological parameters. If the observed statistics fall within the bulk of this distribution, the data are consistent with ΛCDM at the

Table 3: Statistical summary for the MCMC analysis of model `base_CamSpecHM_TT_lowl_lowE`. Columns give the experimental value, MCMC median, 68% credible interval, and empirical p-value.

Statistic	Experimental	MCMC Median	68% CI	p-value
C_{180}	-354.1497 ± 171.2834	261.6952	$[-39.0156, 589.9108]$	0.018
$S_{180,60}$	6694.0882 ± 8822.2915	33937.6161	$[10094.0365, 102450.9702]$	0.076
$S_{180,154}$	2739.9435 ± 3018.8131	5200.7872	$[664.4140, 23292.6452]$	0.368
$S_{154,92}$	2298.0174 ± 3103.8229	10088.0224	$[2572.3400, 34735.6347]$	0.138
$S_{92,30}$	8229.1759 ± 10643.1270	23088.7680	$[9244.2759, 60281.6008]$	0.137
$S_{30,0}$	$39257.1300 \pm 48919.8333$	72409.8155	$[41887.8294, 128755.4170]$	0.133
$\xi_{180,60}$	-2.1052 ± 12.1260	-37.1448	$[-77.0816, -13.6870]$	0.962
$\xi_{180,154}$	-138.6536 ± 57.5605	210.0608	$[-7.9407, 473.6089]$	0.047
$\xi_{154,92}$	42.0072 ± 34.6449	8.7805	$[-22.1704, 48.5205]$	0.802
$\xi_{92,30}$	-81.0404 ± 79.0240	-126.3688	$[-182.5317, -81.9472]$	0.844
$\xi_{30,0}$	377.5806 ± 351.5645	603.5009	$[414.7549, 861.6303]$	0.116



(a) S^{180}_{60} (classic $S_{1/2}$)

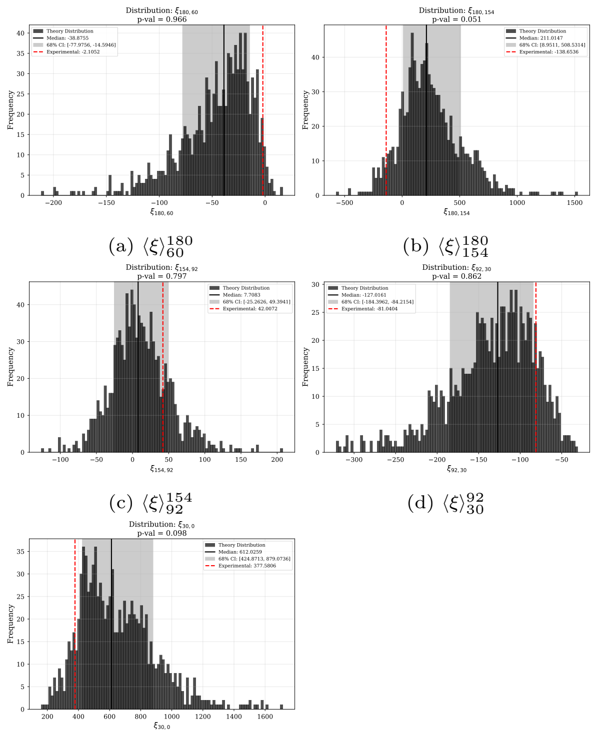
(b) S^{180}_{154}

(c) S^{154}_{92}

(d) S^{92}_{30}

(e) S^{30}_{0}

Figure 4: MCMC posterior distributions of the S^b_a statistic for model `base_CamSpecHM_TT_lowl_lowE`. Each panel shows the distribution over 1000 posterior samples from the Planck CamSpec chain.



(a) $\langle\xi\rangle^{180}_{60}$

(b) $\langle\xi\rangle^{180}_{154}$

(c) $\langle\xi\rangle^{154}_{92}$

(d) $\langle\xi\rangle^{92}_{30}$

(e) $\langle\xi\rangle^{30}_{0}$

Figure 5: MCMC posterior distributions of $\langle\xi\rangle^b_a$ for model `base_CamSpecHM_TT_lowl_lowE`.

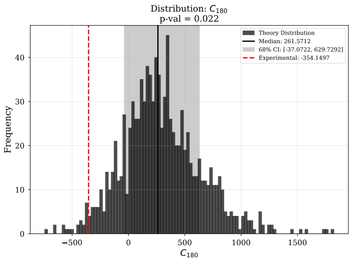


Figure 6: MCMC posterior distribution of C_{180} for model `base_CamSpecHM_TT_lowl_lowE`.

best-fit point and any apparent anomaly is simply a statistical fluctuation of the measured sky.

The MCMC ensemble additionally folds in parameter uncertainty. The broader spread of the MCMC distributions reflects the fact that different points in the posterior predict somewhat different large-angle correlation functions. When the observed statistics fall in the tails of the MCMC-derived distributions, this indicates that the data are not well described by the bulk of the posterior, suggesting a tension between the large-scale and small-scale information encoded in the Planck data.

A key caveat is the look-elsewhere effect: eleven correlated statistics are examined simultaneously, so finding one or more with low p-values is expected even under the null hypothesis. A proper assessment of global significance requires a multivariate test statistic, which is reported in the table summaries. An additional concern at low multipoles is the strong influence of cosmic variance, which means that even a statistic with a p-value of a few per cent does not necessarily signal new physics.

The results for alternative likelihoods and models with free curvature Ω_k are reported in Appendix~Appendix A. The dependence of the results on the likelihood choice (CamSpec vs. Plik) provides a useful cross-check, while the curvature-extended models test whether a departure from flatness could alleviate the observed tensions.

7. Conclusions

We have studied the large-angle CMB angular correlation function and its derived statistics using Planck 2018 data and two complementary ensembles for the theoretical sampling distributions. The statistics examined — the quadratic S_a^b, the linear $\langle\xi\rangle_a^b$, and the antipodal C_{180} — probe distinct aspects of the large-angle temperature field.

The best-fit and MCMC ensembles yield qualitatively different assessments of significance for some statistics, underscoring that the choice of theoretical ensemble matters when evaluating low-ℓ anomalies. A complete summary of all p-values and multivariate tension estimates for the primary `base_CamSpecHM_TT_lowl_lowE` model is provided in Tables~2 and 3; results for additional models appear in Appendix~Appendix A.

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Appendix A. Additional Models: Statistical Tables

The tables in this appendix report the full statistical summary for three additional Planck models. All quantities are defined as in the main text. Model names follow the Planck convention: `omegak` denotes a run with free spatial curvature; `plikHM` denotes the Plik high-multipole likelihood; `TTTEEE` denotes a joint temperature and polarization fit.

Appendix A.1. MCMC Results

Appendix A.2. Best-Fit Results

Table A.4: MCMC results for `base_plikHM_TT_lowl_lowE`.

Statistic	Experimental	MCMC Median	68% CI	p-value
C_{180}	-354.1497 ± 171.2834	241.2170	$[-55.1699, 594.4383]$	0.029
$S_{180,60}$	6694.0882 ± 8822.2915	33801.8449	$[11110.7265, 102182.3499]$	0.065
$S_{180,154}$	2739.9435 ± 3018.8131	4766.0801	$[672.8630, 22097.9814]$	0.376
$S_{154,92}$	2298.0174 ± 3103.8229	10527.3578	$[2848.7506, 34955.3646]$	0.125
$S_{92,30}$	8229.1759 ± 10643.1270	23310.8042	$[9398.8164, 59071.8598]$	0.128
$S_{30,0}$	$39257.1300 \pm 48919.8333$	73229.9679	$[42537.9015, 125384.8849]$	0.126
$\xi_{180,60}$	-2.1052 ± 12.1260	-38.5341	$[-76.0073, -14.4676]$	0.962
$\xi_{180,154}$	-138.6536 ± 57.5605	202.3885	$[-6.9888, 468.8826]$	0.052
$\xi_{154,92}$	42.0072 ± 34.6449	9.1633	$[-25.4628, 51.1022]$	0.781
$\xi_{92,30}$	-81.0404 ± 79.0240	-126.3941	$[-178.6658, -84.8224]$	0.860
$\xi_{30,0}$	377.5806 ± 351.5645	607.6992	$[416.0714, 851.9444]$	0.092

Table A.5: MCMC results for `base_omegak_CamSpecHM_TT_lowl_lowE`.

Statistic	Experimental	MCMC Median	68% CI	p-value
C_{180}	-354.1497 ± 171.2834	208.9942	$[-34.4807, 486.2964]$	0.011
$S_{180,60}$	6694.0882 ± 8822.2915	21765.8299	$[8109.5135, 74119.3517]$	0.119
$S_{180,154}$	2739.9435 ± 3018.8131	3267.8897	$[391.6000, 15964.8461]$	0.466
$S_{154,92}$	2298.0174 ± 3103.8229	7021.0369	$[2079.4052, 26823.1748]$	0.186
$S_{92,30}$	8229.1759 ± 10643.1270	15837.8991	$[6161.9693, 41989.8915]$	0.246
$S_{30,0}$	$39257.1300 \pm 48919.8333$	51438.9870	$[29716.1148, 94216.3869]$	0.310
$\xi_{180,60}$	-2.1052 ± 12.1260	-30.7119	$[-64.0166, -12.3069]$	0.957
$\xi_{180,154}$	-138.6536 ± 57.5605	164.2943	$[-0.5697, 396.1609]$	0.030
$\xi_{154,92}$	42.0072 ± 34.6449	6.7193	$[-20.7478, 41.0697]$	0.847
$\xi_{92,30}$	-81.0404 ± 79.0240	-105.5266	$[-152.8439, -67.5677]$	0.734
$\xi_{30,0}$	377.5806 ± 351.5645	502.4040	$[340.1959, 725.3792]$	0.232

Table A.6: MCMC results for `base_omegak_plikHM_TTTEEE_lowl_lowE`.

Statistic	Experimental	MCMC Median	68% CI	p-value
C_{180}	-354.1497 ± 171.2834	208.6258	$[-45.5817, 506.9834]$	0.013
$S_{180,60}$	6694.0882 ± 8822.2915	25836.5779	$[8014.3153, 74173.0515]$	0.111
$S_{180,154}$	2739.9435 ± 3018.8131	3617.6754	$[464.5482, 16839.1825]$	0.455
$S_{154,92}$	2298.0174 ± 3103.8229	7418.2542	$[2223.0737, 24954.1641]$	0.172
$S_{92,30}$	8229.1759 ± 10643.1270	18231.4590	$[6845.5271, 42367.6853]$	0.205
$S_{30,0}$	$39257.1300 \pm 48919.8333$	55436.6264	$[32260.9180, 96229.5173]$	0.264
$\xi_{180,60}$	-2.1052 ± 12.1260	-33.4438	$[-63.2413, -11.6939]$	0.961
$\xi_{180,154}$	-138.6536 ± 57.5605	168.6679	$[-15.5933, 403.9776]$	0.040
$\xi_{154,92}$	42.0072 ± 34.6449	9.8408	$[-20.0036, 45.5296]$	0.816
$\xi_{92,30}$	-81.0404 ± 79.0240	-110.8947	$[-155.6151, -70.9814]$	0.766
$\xi_{30,0}$	377.5806 ± 351.5645	522.1450	$[353.3445, 736.3491]$	0.196

Table A.7: Best-fit results for `base_plikHM_TT_lowl_lowE`.

Statistic	Experimental	MCMC Median	68% CI	p-value
C_{180}	-354.1497 ± 171.2834	235.3880	$[-62.2566, 586.2191]$	0.028
$S_{180,60}$	6694.0882 ± 8822.2915	33261.4972	$[10608.9403, 100001.9703]$	0.082
$S_{180,154}$	2739.9435 ± 3018.8131	4638.8774	$[579.8045, 21964.4118]$	0.379
$S_{154,92}$	2298.0174 ± 3103.8229	9746.7383	$[2727.4077, 35795.7064]$	0.130
$S_{92,30}$	8229.1759 ± 10643.1270	23100.4791	$[9000.3718, 60298.5831]$	0.145
$S_{30,0}$	$39257.1300 \pm 48919.8333$	70442.7090	$[40918.0369, 130962.8091]$	0.137
$\xi_{180,60}$	-2.1052 ± 12.1260	-37.0548	$[-75.1144, -13.0243]$	0.959
$\xi_{180,154}$	-138.6536 ± 57.5605	195.4772	$[-25.0376, 460.2276]$	0.061
$\xi_{154,92}$	42.0072 ± 34.6449	10.7736	$[-22.3436, 56.5905]$	0.774
$\xi_{92,30}$	-81.0404 ± 79.0240	-124.4639	$[-185.5235, -80.1492]$	0.835
$\xi_{30,0}$	377.5806 ± 351.5645	595.2820	$[409.3589, 866.2165]$	0.113

Table A.8: Best-fit results for `base_omegak_CamSpecHM_TT_lowl_lowE`.

Statistic	Experimental	MCMC Median	68% CI	p-value
C_{180}	-354.1497 ± 171.2834	211.9556	$[-29.8411, 502.2334]$	0.014
$S_{180,60}$	6694.0882 ± 8822.2915	23719.2671	$[7675.5717, 73031.1322]$	0.131
$S_{180,154}$	2739.9435 ± 3018.8131	3625.1233	$[406.7749, 15623.4577]$	0.453
$S_{154,92}$	2298.0174 ± 3103.8229	7128.0973	$[1943.9653, 24772.1131]$	0.209
$S_{92,30}$	8229.1759 ± 10643.1270	16340.3031	$[6726.9125, 41856.6421]$	0.223
$S_{30,0}$	$39257.1300 \pm 48919.8333$	51705.4424	$[31338.3396, 91825.7121]$	0.295
$\xi_{180,60}$	-2.1052 ± 12.1260	-32.3840	$[-63.4182, -11.6872]$	0.953
$\xi_{180,154}$	-138.6536 ± 57.5605	172.3367	$[-0.9287, 394.2349]$	0.034
$\xi_{154,92}$	42.0072 ± 34.6449	8.3009	$[-20.7864, 42.4205]$	0.835
$\xi_{92,30}$	-81.0404 ± 79.0240	-105.6599	$[-151.8632, -71.0518]$	0.747
$\xi_{30,0}$	377.5806 ± 351.5645	504.0592	$[354.5585, 720.7156]$	0.211

Table A.9: Best-fit results for `base_omegak_plikHM_TTTEEE_lowl_lowE`.

Statistic	Experimental	MCMC Median	68% CI	p-value
C_{180}	-354.1497 ± 171.2834	218.3127	$[-17.0392, 508.2853]$	0.006
$S_{180,60}$	6694.0882 ± 8822.2915	24349.6721	$[7615.2117, 72133.8665]$	0.136
$S_{180,154}$	2739.9435 ± 3018.8131	3289.8516	$[501.4208, 16347.5457]$	0.459
$S_{154,92}$	2298.0174 ± 3103.8229	7095.9810	$[2120.8351, 26428.7002]$	0.171
$S_{92,30}$	8229.1759 ± 10643.1270	17049.6488	$[6261.6264, 42388.9107]$	0.224
$S_{30,0}$	$39257.1300 \pm 48919.8333$	54447.2715	$[31086.6010, 96125.6258]$	0.280
$\xi_{180,60}$	-2.1052 ± 12.1260	-33.5933	$[-63.5443, -11.2915]$	0.967
$\xi_{180,154}$	-138.6536 ± 57.5605	172.7077	$[9.0572, 400.9675]$	0.020
$\xi_{154,92}$	42.0072 ± 34.6449	7.2581	$[-21.0722, 40.1496]$	0.853
$\xi_{92,30}$	-81.0404 ± 79.0240	-106.5318	$[-154.6407, -69.6381]$	0.744
$\xi_{30,0}$	377.5806 ± 351.5645	518.2218	$[349.5821, 732.1247]$	0.211